Class 10 Math - Chapter 8: Introduction to Trigonometry

🌟 Introduction

Trigonometry is the study of relationships between the sides and angles of a triangle. The word comes from Greek: *tri* (three), *gon* (sides), and *metron* (measure).

In Class 10, we strictly focus on **Right-Angled Triangles**.

The Sides of a Right Triangle:

For a given acute angle θ:

- **Hypotenuse (H):** The longest side, directly opposite the 90° angle.
- **Perpendicular/Opposite (P):** The side directly opposite to angle θ.
- **Base/Adjacent (B):** The side that forms angle θ alongside the hypotenuse.

📌 Trigonometric Ratios

There are 6 fundamental trigonometric ratios that compare the lengths of two sides of the right triangle.

> [!NOTE] Mnemonic to remember the core three: **SOH CAH TOA**
>
> - Sin = Opposite / Hypotenuse
> - Cos = Adjacent / Hypotenuse
> - Tan = Opposite / Adjacent

Ratio	Formula	Reciprocal Ratio	Formula
Sine ($\sin \theta$)	P / H	Cosecant ($\csc \theta$)	H / P
Cosine ($\cos \theta$)	B / H	Secant ($\sec \theta$)	H / B
Tangent ($\tan \theta$)	P / B	Cotangent ($\cot \theta$)	B / P

> [!TIP] Tangent can also be expressed directly as the ratio of sine to cosine: $\tan \theta = \frac{\sin \theta}{\cos \theta}$

⚡ Trigonometric Identities

These identities hold true for absolutely ALL values of angle A:

1. $\sin^2 A + \cos^2 A = 1$
2. $1 + \tan^2 A = \sec^2 A$
3. $1 + \cot^2 A = \csc^2 A$

Standard Angles Table

Angle (θ)	0°	30°	45°	60°
$\sin \theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$
$\cos \theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$

(Notice how the purely values of $\cos$ are exactly the values of $\sin$ written smoothly in reverse order!)

❌ Common Mistakes

> [!WARNING]
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> 1. **Squares Notation:** The square of $\sin \theta$ is correctly written as $\sin^2 \theta$. Writing it as $\sin \theta^2$ implies that ONLY the angle θ is being squared, which is mathematically completely different!
> 2. **Base vs. Perpendicular:** The base and perpendicular SWAP depending on which acute angle you are looking at. Always trace from the specific angle in question!
> 3. **Assuming Proportionality means Equality:** If $\sin A = 3/5$, it doesn't mean $P=3$ cm and $H=5$ cm. It means $P=3k$ and $H=5k$, where k is a constant. Always use k in length problems!

📝 Quick Revision

Summary

Trigonometry binds algebra and geometry by translating angle degrees into side-length ratios. Mastering the SOH-CAH-TOA relationships, memorizing the $0-90^\circ$ standard angle values perfectly, and fluently rotating through the three main square identities will allow you to solve any proof or height-distance problem!